

Technology Stack

Date: 29 September 2026	Team ID: SWTID-2026-5955
Project Name: PocketSmart AI - Your Smart Budget & Recommendation Assistant	Maximum Marks: 2 Marks

Define Your Technology Stack

PocketSmart AI is an AI-powered budget and recommendation assistant designed to help users monitor spending, plan budgets for home essentials, parties, and jewelry purchases, and receive intelligent recommendations. The technology stack is designed to be lightweight, scalable, and efficient, enabling rapid AI-driven recommendation generation while maintaining responsive web performance and ease of deployment for student and consumer use cases. The stack leverages Python web frameworks for backend processing, integrates with generative AI APIs for recommendation tasks, and uses responsive front-end technologies to ensure accessibility on mobile and desktop for budget planning.

Technology Stack Details

S.No.	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (React, Angular, HTML/CSS)	HTML5, CSS3, JavaScript, Jinja2 Templates	- Lightweight and responsive, easy integration with Flask/FastAPI - Jinja2 enables dynamic rendering of AI recommendation - Mobile-friendly design for budget planners, fast loading
2	Backend / Server-Side (Node.js, Python/Django, Java)	Python with Flask & FastAPI Framework	- Fast processing of AI requests - Easy integration with Gemini 1.5 Flash Pro API - Modular routing: /generate-home, /generate-party, /generate-jewelry - Existing team expertise in Python, async support



Technology Stack Continuation

Continuation of Technology Stack details from Page 1 — Components 3 to 5

No.	Component	Technology Chosen	Justification
3	 Database / Data Storage (MySQL, MongoDB, PostgreSQL)	SQLite for development, PostgreSQL / IBM Cloud Database for production, Flask-Session for session storage.	Lightweight SQLite for rapid prototyping; PostgreSQL handles structured user data, recommendation history, secure session handling via /session-info /session-data; strong relational mapping; scalable.
4	 Cloud / Hosting / Deployment (AWS, Azure, Heroku, Docker)	IBM Cloud, Docker, Uvicorn Server.	Easy scalability; Docker containerization for consistent deployment; IBM Cloud supports Watsonx and Python apps; Uvicorn for high-performance FastAPI serving; cost-effective hosting for student project.
5	 Version Control & CI/CD (GitHub, GitLab, Jenkins)	GitHub, Git.	Seamless collaboration among team; version tracking for gemini_uutls.py and routes; GitHub Actions for automated testing; easy rollback; code review.



Summary

- The stack prioritizes scalability, reliability, and cost-effectiveness for PocketSmart AI.
- SQLite + PostgreSQL ensures smooth transition from development to production.
- IBM Cloud & Docker enable cloud-ready deployment aligned with Watsonx integration.
- GitHub & CI/CD maintain code quality and collaboration for the project.

AI-Powered Budget Shopping Assistant • Version 1.0 • Date: 2024-12

6. Third-Party APIs / Other Tools (Stripe, Twilio, Google Maps)



Row	Technology Chosen	Justification
6	• Google Gemini 1.5 Flash Pro API • Amazon Product Advertising API • IKEA API • Flipkart Affiliate • Swiggy / Zomato APIs • OYO API • CORS • JWT	Gemini 1.5 Flash Pro enables multimodal AI (text + image) for budget-aware recommendations across Home / Party / Jewelry, processes outfit images for jewelry matching. E-commerce APIs enable real-time product sourcing. JWT ensures secure auth. CORS enables frontend-backend secure communication. Provides personalized, budget-friendly suggestions.

Key Benefits:

- Personalized recommendations tailored to user budget & style preferences
- Scalable integration with major e-commerce and service providers
- Secure and reliable data communication via JWT & CORS